GRIDGUARD AI 4.0

National Smart Grid Executive Intelligence Report

Generated: 2026-02-09T04:32:02.979Z
Scope: National operational risk, theft intelligence, transformer reliability, alert posture, and recommended actions.
Source: Live platform data (regions, alerts, transformer telemetry, meter anomalies, and reading streams).

Executive Narrative

This executive brief presents a formal live operational assessment of GRIDGUARD AI 4.0 based on current telemetry, anomaly outputs, and alert intelligence.

At the time of generation, national average theft score stands at 21.31% and national distribution loss stands at 7.17%. The transformer risk index is 44.29% and the carbon optimization score is 70.43%.

System footprint currently includes 35 regions, 120 transformers, and 1080 smart meters.

Risk zoning currently indicates 0 red zones, 1 yellow zones, and 34 green zones.

The estimated daily financial impact of loss is INR 28,201 based on recent load behavior and live loss conditions.

Executive Summary:
- National loss remains elevated and requires feeder-level loss reduction focus.
- Transformer risk is moderate with pockets of overload stress.
- Theft exposure is concentrated in a subset of high-density regions.
- Prioritize vigilance dispatch, load balancing, and preventive transformer maintenance.

Key Metrics

National Theft Score (Avg): 21.31%
National Distribution Loss (Avg): 7.17%
Transformer Risk (Avg): 44.29%
Carbon Optimization (Avg): 70.43%
Open Alerts: 123
High Alerts (24h): 3
Estimated Daily Financial Impact (INR): 28,201

National Operational Snapshot

Total Regions Monitored: 35
Total Transformers Monitored: 120
Total Smart Meters Monitored: 1080
National Theft Score (Avg): 21.31%
National Distribution Loss (Avg): 7.17%
National Transformer Risk (Avg): 44.29%
National Carbon Optimization Score (Avg): 70.43%

Risk Zone Distribution

Red Zones (>70 theft score): 0
Yellow Zones (40-70 theft score): 1
Green Zones (<40 theft score): 34

Alert Intelligence

Total Alerts Logged: 131
Open Alerts (Unacknowledged): 123
High Severity Alerts in Last 24h: 3

- [2026-02-09T04:28:32.312Z] (MEDIUM) Delhi: Anomaly detected for meter 213bce00-b18f-4ba1-b67f-9d26e69c56d5
- [2026-02-09T04:28:23.268Z] (MEDIUM) Tamil Nadu: Anomaly detected for meter e3e09d2c-5454-4771-a5b9-4cd3f367387e
- [2026-02-08T06:59:34.922Z] (MEDIUM) Maharashtra: Anomaly detected for meter d1eb1964-df64-46d8-86bd-114a0451928f
- [2026-02-08T06:59:04.759Z] (MEDIUM) Delhi: Anomaly detected for meter 8cca7c5e-4b43-4406-a410-59eb8d6afb92
- [2026-02-08T06:58:52.689Z] (MEDIUM) Delhi: Anomaly detected for meter 582a0594-17eb-4f71-95ad-95c602685831
- [2026-02-08T06:58:43.638Z] (MEDIUM) Karnataka: Anomaly detected for meter d5fa83ba-cc64-4320-83d9-ff2810edae74
- [2026-02-08T06:58:40.623Z] (MEDIUM) Bihar: Anomaly detected for meter 94ca19e9-46f6-4855-b709-df8df4b4a861
- [2026-02-08T06:58:31.572Z] (HIGH) Maharashtra: Anomaly detected for meter 23849cd3-abef-4eee-a2c7-e89b2f3c2069
- [2026-02-08T06:58:22.519Z] (MEDIUM) Gujarat: Anomaly detected for meter ef5866ac-fb68-44f0-998e-2e56b2b3d54e
- [2026-02-08T06:58:07.427Z] (HIGH) Karnataka: Anomaly detected for meter 7ee61712-2072-4bb3-84f8-d7f69f0b12e7
- [2026-02-08T06:57:49.317Z] (MEDIUM) Karnataka: Anomaly detected for meter 9b02c610-18c8-4fed-9f5d-4149e6e7b807
- [2026-02-08T06:56:12.788Z] (MEDIUM) Tamil Nadu: Anomaly detected for meter 408f454e-47c4-47b4-a844-230050a8a61d

Top High-Risk Regions

- 1. Chandigarh | Theft 65% | Loss 21.76% | Transformer Risk 54.82% | Carbon 70% | Updated 2026-02-08T06:55:45.651Z
- 2. Mizoram | Theft 34% | Loss 11.29% | Transformer Risk 43.27% | Carbon 59% | Updated 2026-02-07T22:17:27.901Z
- 3. West Bengal | Theft 34% | Loss 6.71% | Transformer Risk 41.29% | Carbon 73% | Updated 2026-02-07T22:17:27.901Z
- 4. Sikkim | Theft 33% | Loss 5.17% | Transformer Risk 48.68% | Carbon 85% | Updated 2026-02-07T22:17:27.901Z
- 5. Himachal Pradesh | Theft 32% | Loss 9.67% | Transformer Risk 27.63% | Carbon 64% | Updated 2026-02-07T22:17:27.901Z
- 6. Dadra and Nagar Haveli | Theft 30% | Loss 4.91% | Transformer Risk 23.51% | Carbon 62% | Updated 2026-02-07T22:17:27.901Z
- 7. Jharkhand | Theft 29% | Loss 6.51% | Transformer Risk 45.79% | Carbon 67% | Updated 2026-02-07T22:17:27.901Z

- 8. Andhra Pradesh | Theft 27% | Loss 8.91% | Transformer Risk 72.84% | Carbon 70% | Updated 2026-02-09T04:30:57.029Z
- 9. Madhya Pradesh | Theft 26% | Loss 10.36% | Transformer Risk 34.43% | Carbon 70% | Updated 2026-02-07T22:17:27.901Z
- 10. Orissa | Theft 25% | Loss 7.43% | Transformer Risk 63.02% | Carbon 67% | Updated 2026-02-07T22:17:27.901Z

Top Vulnerable Transformers

- 1. Delhi | Transformer d232ecfb | Load 74.48% | Temp 54.1 C | Health 0.38 | Risk GREEN
- 2. Maharashtra | Transformer 7fd2640e | Load 73.83% | Temp 62.88 C | Health 0.38 | Risk GREEN
- 3. Andhra Pradesh | Transformer 9ab80d8c | Load 72.84% | Temp 68.3 C | Health 0.39 | Risk GREEN
- 4. Karnataka | Transformer f0f0fe44 | Load 72.18% | Temp 63.72 C | Health 0.4 | Risk GREEN
- 5. Tamil Nadu | Transformer df0db96a | Load 72.09% | Temp 50.65 C | Health 0.4 | Risk GREEN
- 6. Tamil Nadu | Transformer f833d1ba | Load 71.9% | Temp 46.42 C | Health 0.4 | Risk GREEN
- 7. Bihar | Transformer bb9efdb5 | Load 71.68% | Temp 47.72 C | Health 0.4 | Risk GREEN
- 8. Arunachal Pradesh | Transformer 9b38e0e9 | Load 71.66% | Temp 49.14 C | Health 0.4 | Risk GREEN
- 9. Delhi | Transformer 51302981 | Load 71.29% | Temp 49.83 C | Health 0.41 | Risk GREEN
- 10. Chandigarh | Transformer 1992130b | Load 70.39% | Temp 50.57 C | Health 0.41 | Risk GREEN

Top Meter Anomalies (7-day window)

- 1. Arunachal Pradesh | Meter 13821bee-947 | Theft Probability 84% | Class RED | Detected 2026-02-07T23:20:42.183Z
- 2. Gujarat | Meter 81ba998c-1b0 | Theft Probability 83% | Class RED | Detected 2026-02-07T23:26:38.082Z
- 3. Delhi | Meter 9657afa8-a5b | Theft Probability 83% | Class RED | Detected 2026-02-07T21:50:38.095Z
- 4. Delhi | Meter 70214083-d40 | Theft Probability 82% | Class RED | Detected 2026-02-07T23:43:33.695Z
- 5. Gujarat | Meter 375a665d-c07 | Theft Probability 82% | Class RED | Detected 2026-02-07T22:36:28.438Z
- 6. Karnataka | Meter 7ee61712-207 | Theft Probability 81% | Class RED | Detected 2026-02-08T06:58:07.420Z
- 7. Karnataka | Meter 417f1359-105 | Theft Probability 81% | Class RED | Detected 2026-02-07T22:42:26.999Z
- 8. Tamil Nadu | Meter 98223e50-2a8 | Theft Probability 81% | Class RED | Detected 2026-02-07T22:36:04.297Z
- 9. Maharashtra | Meter d8ef16cf-5fa | Theft Probability 80% | Class RED | Detected 2026-02-07T22:55:52.712Z
- 10. Tamil Nadu | Meter 74331734-0ba | Theft Probability 79% | Class RED | Detected 2026-02-07T23:38:07.588Z

24-Hour Load Trend

Average Power (24h): 2.167 kW
Estimated Daily Energy Loss: 4028.71 kWh

- 2026-02-08T06:30:00.000Z | Avg Power 2.185 kW | Readings 298
- 2026-02-09T03:30:00.000Z | Avg Power 2.13 kW | Readings 28

- 2026-02-09T04:30:00.000Z | Avg Power 2.185 kW | Readings 18

Recommended Actions

- Maintain active anomaly surveillance and sustain monthly anti-theft audit cadence in medium-risk states.
- Current national loss is within an acceptable band; continue preventive maintenance and loss monitoring.
- Initiate a 14-day transformer reliability sprint focused on high-load assets with low health index.
- Establish an alert triage war-room with SLA-based acknowledgement and closure tracking.
- Use LLM assistant for weekly executive briefings, including state-wise risk rationale and recommended field actions.

Methodology Note

This report is generated from live PostgreSQL operational data and near-real-time simulator/ingest streams.
Risk zone classification follows GRIDGUARD thresholds based on theft probability and transformer stress signals.
Financial impact estimation uses current average power, meter coverage, and loss percentage with an indicative INR conversion factor.

Generated by GRIDGUARD AI 4.0 automated reporting engine.